Beyond Differences: Doubly Robust Meta-Learners for Ratio-Based Treatment Effects*

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Abstract

When treatment effects are naturally expressed as ratios — as in medicine, pricing, and marketing, where varying baseline rates make relative change the natural scale — the ratio-based CATE $\tau(x) = \mathbb{E}[Y|W{=}1, X{=}x]/\mathbb{E}[Y|W{=}0, X{=}x]$ is the appropriate estimand. Yet existing estimators either impose log-linear parametric structure or apply generic regression without robustness guarantees for this functional. We introduce the *Q-Learner*, which decomposes $\tau(x)$ into a product of two odds ratios, reducing ratio-CATE estimation for binary outcomes to two propensity classification tasks. We further derive doubly robust augmentations for both S/T- and Q-style ratio learners and characterize their distinct robustness properties. In benchmarks on seven RCT datasets, no single learner dominates, but overall plug-in learners prove sufficient, with the S- and Q-Learner standing out as the most consistent across datasets. On five observational datasets, where propensity must be estimated and confounding cannot be ruled out, the log-scale DR learners introduced here are clearly superior.

1 Introduction

In many domains, treatment effects are naturally expressed as ratios. In medicine, relative effects are the dominant reporting format for both trials and subgroup analyses [Andersen, 2021]. In pricing, treatment effects are most naturally expressed as elasticities — ratios of purchase rates at different price levels — rather than absolute changes in conversion. In marketing, treatment effects are routinely reported as relative lift over baseline conversion. In each case, the ratio-based CATE

$$\tau(x) = \frac{\mathbb{E}[Y|W{=}1, X{=}x]}{\mathbb{E}[Y|W{=}0, X{=}x]} \tag{1}$$

is the appropriate estimand.

The ratio scale is also often more stable than the absolute scale. Schmid et al. [1998] give empirical evidence: control-group event rates predict absolute but not relative treatment effects across clinical trials, suggesting that the ratio scale better captures the intrinsic treatment effect. The same point is visible mechanically: the same absolute effect $\tau_{\text{diff}}(x) = 0.05$ corresponds to a sixfold increase when $\mu_0(x) = 0.01$ but only a 25% lift when $\mu_0(x) = 0.2$. This mechanism is typical when baselines are low and variable — the standard regime in clinical trials (1–20%) and marketing campaigns (1–10%). When baselines are instead high and homogeneous, the difference-based CATE is equally natural and often easier to estimate.

Difference-based CATE estimation

$$\tau_{\text{diff}}(x) = \mathbb{E}[Y|W{=}1, X{=}x] - \mathbb{E}[Y|W{=}0, X{=}x] \tag{2}$$

*Code, data, and scripts to reproduce the experiments are available at https://github.com/michaelfuchs90/ratiobasedcate.

is a mature area, with established meta-learners [Künzel et al., 2019, Nie and Wager, 2021] and doubly robust constructions [Kennedy, 2023]. The literature on ratio-based CATE estimation is much thinner. Existing approaches either impose parametric log-linear structure that cannot capture complex heterogeneity [Yadlowsky et al., 2021, Zou, 2004], or apply generic outcome regression (S- or T-Learner style; Künzel et al., 2019) without exploiting the multiplicative structure of the ratio functional and without robustness guarantees against model misspecification.

One might attempt to recover $\tau(x)$ from existing difference-based estimators via $\tau(x) = 1 + \tau_{\text{diff}}(x)/\mu_0(x)$, but this requires an additional estimate of $\mu_0(x)$, and errors in both $\hat{\tau}_{\text{diff}}$ and $\hat{\mu}_0$ compound through the division—particularly when $\mu_0(x)$ is small, which is precisely the regime where the ratio-based CATE is most valuable. Alternatively, one could turn the ratio into a difference via $\log \tau(x) = \log \mu_1(x) - \log \mu_0(x)$ and apply existing difference-based methods on the log scale. However, this still requires separate log-outcome models, and robustness guarantees derived for difference-based estimators do not transfer through nonlinear transformations. Dedicated methods for the ratio functional are therefore necessary.

1.1 Contributions

For binary outcomes (e.g., conversion or claim occurrence), where $\tau(x)$ coincides with the conditional relative risk, this paper makes two contributions. First, we introduce the *Q-Learner*, which exploits a Bayes-rule identity to decompose the ratio CATE into a product of two odds ratios:

$$\tau(x) = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)},$$

where $p(x) = P(W{=}1|Y{=}1, X{=}x)$ is the converter propensity and $e(x) = P(W{=}1|X{=}x)$ is the treatment propensity. This reduces ratio CATE estimation to two binary classification tasks (a propensity model and a converter-propensity model) without requiring outcome regression. At low conversion rates, this also converts a heavily imbalanced regression task into two balanced classification tasks and sidesteps the numerical instability of $\hat{\mu}_0$ in the denominator. The Q-Learner directly optimizes $\hat{\tau}(x)$ through its component classifiers, making it naturally amenable to hyperparameter tuning and model selection. We further show that estimating propensity from data is preferable to using the known design propensity, even in RCTs. We introduce the Q-Simple Learner to confirm this empirically.

Second, we derive *doubly robust augmentations* for both S/T-style and Q-style ratio learners, on both direct and log scale, using influence-function-based corrections. These augmentations eliminate first-order bias from nuisance estimation errors, providing formal robustness guarantees. The two constructions exhibit *different* robustness properties: DR-S/T achieves classical double robustness (consistency if either propensity or outcome model is correct), while DR-Q achieves only conditional robustness requiring exact propensity knowledge—a distinction with empirical consequences in observational settings.

We evaluate all methods on seven RCT and five observational datasets (100 seeds each). In experimental settings with known propensity, no single learner dominates, but overall plug-in learners prove sufficient, with the S- and Q-Learner standing out as the most consistent across datasets. In observational settings, where propensity must be estimated and confounding cannot be ruled out, the S/T log-scale DR learners introduced here are clearly superior, confirming the importance of classical double robustness in such settings.

1.2 Related Work

Meta-Learners and Doubly Robust Foundations. Difference-based CATE methodology relies on the meta-learner paradigm of Künzel et al. [2019] and Nie and Wager [2021], and on the pseudo-outcome and influence-function machinery developed in Robins et al. [1994], Chernozhukov et al. [2018], Kennedy [2023], and Foster and Syrgkanis [2023]. Our DR-S/T and DR-Q constructions adapt this machinery to the ratio functional.

Ratio-based CATE Estimation. Closest to our setting, Yadlowsky et al. [2021] propose a doubly robust estimator for the ratio-based CATE restricted to a log-linear semiparametric model $D(z) = \exp(\beta_0^\top z)$. Our work imposes no parametric structure on $\tau(x)$, introduces the Q-Learner parameterisation that requires no outcome regression, and derives two distinct DR augmentations with differing robustness properties. Causal forests [Athey et al., 2019] estimate heterogeneous effects via local moment conditions and can in principle target non-standard functionals, but do not provide an out-of-the-box estimator or efficiency theory for the ratio CATE.

Relative Risk Estimation. Log-binomial and modified Poisson regression [Zou, 2004] estimate marginal or conditionally linear relative risks but cannot capture nonlinear heterogeneity. Richardson et al. [2017] develop a semiparametric framework for the joint modelling of the relative risk and the risk difference, characterising the nuisance tangent space for these functionals. Shirvaikar et al. [2023] modify causal forests to target relative risk via GLM-based node splitting; their approach retains a GLM-based splitting criterion and does not provide doubly robust guarantees. Boughdiri et al. [2025] derive doubly robust estimators for the marginal risk ratio, establishing asymptotic normality and confidence intervals; their work targets population-level effects rather than heterogeneous CATEs. Concurrently, Gao and Hastie [2025] propose the DINA estimator (difference in natural parameters) via an R-Learner-style extension for exponential families. For Bernoulli outcomes DINA targets the conditional log-odds ratio; for Poisson (count) outcomes it coincides with the conditional log risk ratio, paralleling our log-scale variants. Their method requires choosing an outcome family, whereas the Q-Learner identity and our DR augmentations target the risk ratio directly without such a specification. DINA also differs in construction: it adapts the R-Learner's Robinson-style loss via exponential-family likelihoods, whereas our DR learners use explicit influence-function-based pseudo-outcomes.

Uplift Modeling. The marketing literature [Radcliffe and Surry, 2011] focuses on ranking by treatment effect. We address both ranking and calibration.

The remainder of this paper is organized as follows. Section 2 establishes notation and assumptions. Section 3 derives the Q-Learner identity. Section 4 develops the doubly robust extensions. Section 5 presents the empirical evaluation, and Section 6 concludes.

2 Setup and Notation

We work within the standard potential outcomes framework [Rubin, 1974]. Each unit is characterized by a vector of pre-treatment covariates $X \in \mathcal{X}$, a binary treatment indicator $W \in \{0, 1\}$, and a binary outcome $Y \in \{0, 1\}$.

Our target estimand is the ratio-based conditional average treatment effect:

$$\tau(x) = \frac{\mu_1(x)}{\mu_0(x)}, \tag{3}$$

where $\mu_w(x) = P(Y{=}1 \mid W{=}w, X{=}x)$ denotes the response surface under treatment arm w. The ratio $\tau(x)$ quantifies how much more likely the outcome becomes under treatment relative to control, for units with covariates x.

Identification of $\tau(x)$ from observational data requires two standard assumptions:

Assumption 1 (Unconfoundedness)**.** $(Y(1), Y(0)) \perp W \mid X$

Assumption 2 (Overlap). $0 < e(x) < 1$ *for all* $x \in \mathcal{X}$*,*

where $e(x) = P(W{=}1 \mid X{=}x)$ is the propensity score. In addition, the ratio functional requires that the denominator is bounded away from zero:

Assumption 3 (Positivity of Control Response). $\mu_0(x) > 0$ *for all* $x \in \mathcal{X}$*.*

This assumption ensures that $\tau(x)$ is well-defined and is satisfied whenever the outcome has non-negligible baseline prevalence—the typical case in the applications we consider (conversion rates $\geq 1\%$). In practice, regions where $\mu_0(x)$ is very small lead to unstable estimates; we address this through clipping (Appendix B).

Beyond these standard quantities, the Q-Learner (Section 3) relies on two additional objects:

- The **converter propensity** $p(x) = P(W{=}1 \mid Y{=}1, X{=}x)$: the probability of having been treated, conditional on a positive outcome.
- The **marginal conversion probability** $m(x) = P(Y{=}1 \mid X{=}x)$: the unconditional outcome rate given covariates.

3 The Q-Learner

We now derive a meta-learner that directly exploits the multiplicative structure of the ratio CATE. The key observation is the following identification result:

Under Assumptions 1–3, for binary outcomes $Y \in \{0, 1\}$:

$$\tau(x) = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)} \tag{4}$$

where $p(x) = P(W{=}1|Y{=}1, X{=}x)$ is the treatment probability among converters and $e(x) = P(W{=}1|X{=}x)$ is the propensity. To prove the relation, apply Bayes' theorem to $\mu_1(x) = P(Y{=}1|W{=}1, X{=}x)$:

$$\mu_1(x) = \frac{P(W{=}1|Y{=}1, X{=}x) \cdot P(Y{=}1|X{=}x)}{P(W{=}1|X{=}x)} = \frac{p(x) \cdot m(x)}{e(x)} \tag{5}$$

Similarly, $\mu_0(x) = (1 - p(x)) \cdot m(x)/(1 - e(x))$. Taking the ratio, $m(x)$ cancels:

$$\tau(x) = \frac{\mu_1(x)}{\mu_0(x)} = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)} \tag{6}$$

The identity has an intuitive interpretation: if treatment increases conversion ($\tau(x) > 1$), then converters are disproportionately likely to have been treated relative to the overall propensity, i.e., $p(x) > e(x)$. The ratio $\frac{p(x)/(1-p(x))}{e(x)/(1-e(x))}$ quantifies exactly this excess on the odds scale.

The Q-Learner identity suggests a simple estimation strategy:

1. **Step 1:** Estimate propensity score $\hat{e}(x)$ from the observed treatment assignments using all observations (Section 3.1 explains why estimation is preferred even when $e(x)$ is known)
2. **Step 2:** Estimate converter treatment probability $\hat{p}(x)$ using only observations with $Y = 1$
3. **Step 3:** Compute CATE estimate: $\hat{\tau}(x) = \frac{\hat{p}(x)}{1-\hat{p}(x)} \cdot \frac{1-\hat{e}(x)}{\hat{e}(x)}$

Compared to S- and T-Learners, the Q-Learner offers three structural advantages:

- **Direct optimization of τ**: Since $\hat{\tau}(x)$ is a deterministic function of $\hat{p}(x)$ and $\hat{e}(x)$, hyperparameter tuning of the component models directly improves the treatment effect estimate. In contrast, the S-Learner optimizes the outcome model $\hat{\mu}(x, w)$, and improvements in outcome prediction do not necessarily translate to better treatment effect estimation. This makes the Q-Learner particularly amenable to automated model selection (Section 5).
- **Balanced classification**: In A/B tests with $e(x) \approx 0.5$ and low conversion rates, the Q-Learner transforms a highly imbalanced outcome regression problem into a balanced classification problem among converters.
- **Flexibility**: Any probabilistic classifier can serve as base model for both stages.

Remark 1 (Single Model via Offset)**.** *Taking logs of* (4) *one arrives at*

$$\log \tau(x) = \mathit{logit}(p(x)) - \mathit{logit}(e(x)). \tag{7}$$

This suggests fitting a single logistic regression for $p(x)$ with $\mathit{logit}(\hat{e}(x))$ as offset:

$$\mathit{logit}(p(x)) = \mathit{logit}(\hat{e}(x)) + f(x) \tag{8}$$

where $f(x)$ models the treatment effect heterogeneity. The fitted model directly yields $\hat{\tau}(x) = \exp(\hat{f}(x))$, reducing the two-stage procedure to a single model after propensity estimation.

Remark 2 (Extension to Multi-Class Treatments)**.** *The Q-Learner can be generalized to non-binary treatments by using multi-class classification for $p(w, x) = P(W{=}w|Y{=}1, X{=}x)$ and $e(w, x) = P(W{=}w|X{=}x)$:*

$$\tau(x, w_1, w_2) = \frac{p(w_1, x)}{p(w_2, x)} \cdot \frac{e(w_2, x)}{e(w_1, x)} \tag{9}$$

Extensions to non-binary outcomes are possible but less straightforward; we leave their investigation to future work.

3.1 Propensity Estimation and the Q-Simple Learner

In RCT settings where propensity is exactly known, one might be tempted to skip fitting a propensity model and use the known $e(x)$ directly in the Q-Learner identity (4). This reduces CATE estimation to a single model for $p(x)$, estimated only among converters. This is a computationally attractive simplification, especially for large datasets.

However, estimating propensity from data can reduce finite-sample variance. The key insight is that $\hat{p}(x)$ and $\hat{e}(x)$ are correlated through the same realized treatment assignments: if a covariate region happens to contain more treated units than expected, both $\hat{e}(x)$ and $\hat{p}(x)$ increase, and the biases partially cancel in the product $\frac{\hat{p}}{1-\hat{p}} \cdot \frac{1-\hat{e}}{\hat{e}}$. Using the known $e(x)$ breaks this correlation, potentially increasing finite-sample variance. Fitting $\hat{e}(x)$ from data—even though the model only captures the randomness of the treatment assignments—preserves this correlation and thereby reduces variance.

This variance reduction diminishes with sample size: as n grows, the realized propensity concentrates around its true value, and the correlation benefit vanishes. For large datasets, the computational savings from skipping propensity estimation may therefore outweigh the small variance cost.

The Q-Simple Learner. To empirically quantify this trade-off, we introduce the *Q-Simple Learner*: a variant that uses the known propensity $e(x)$ directly in the Q-Learner identity (4), without estimating it from data. For RCTs with constant propensity (e.g., $e = 0.5$), this simply means plugging in the design constant. The Q-Simple Learner is only applicable to experimental settings where $e(x)$ is known by design.

Section 5 confirms that Q-Learner outperforms Q-Simple on average (+0.015 mean Qini), though the advantage is concentrated on a single dataset (Lenta: +0.087); on the remaining six RCTs the difference is negligible (+0.003). For large datasets where computational efficiency matters, Q-Simple remains a competitive choice.

4 Doubly Robust Extensions

The Q-Learner and the naive ratio estimator $\hat{\mu}_1(x)/\hat{\mu}_0(x)$ (T-Learner) are plug-in estimators: they combine estimated nuisance functions into $\hat{\tau}(x)$ without correction for estimation error. If any nuisance model is misspecified, bias propagates directly into the CATE estimate.

Following the framework of Kennedy [2022], we derive *augmented* estimators that eliminate the leading-order bias through influence-function-based corrections. Concretely, for an estimand $\tau(x)$ that depends on nuisance functions g (propensity, outcome models, etc.), the augmented pseudo-outcome takes the form

$$\Gamma = \hat{\tau}(x) + \hat{\phi}_\tau(W, Y, X; x), \tag{10}$$

where $\hat{\phi}_\tau$ is the estimated influence function of τ evaluated at the nuisance estimates. This correction removes the first-order (linear) bias term, leaving a remainder of order $O_p(\|\hat{g} - g\|_2^2)$—a product of nuisance estimation errors that vanishes faster than any individual error alone. The resulting estimators are *oracle-efficient* in the sense of Kennedy [2022] when nuisance estimates converge at sufficiently fast rates: regressing the pseudo-outcome Γ on X then achieves the same convergence rate as if the true nuisance functions were known.

The ratio CATE $\tau(x) = \mu_1(x)/\mu_0(x)$ admits two natural parameterizations: through outcome models (μ_1, μ_0) or through the Q-Learner reparameterization (p, e). Each leads to a distinct augmented estimator. As we show below, these exhibit fundamentally different robustness properties: DR-S/T achieves classical double robustness, while DR-Q achieves only conditional robustness requiring exact propensity knowledge.

4.1 DR-S/T: Augmenting the Ratio via Outcome Models

Both estimate $\hat{\tau}(x) = \hat{\mu}_1(x)/\hat{\mu}_0(x)$ as a plug-in ratio and share the following augmentation. The standard AIPW influence functions for the response surfaces are:

$$\phi_{\mu_1}(W, Y, X; x) = \frac{W(Y - \mu_1(x))}{e(x)}, \tag{11}$$

$$\phi_{\mu_0}(W, Y, X; x) = \frac{(1 - W)(Y - \mu_0(x))}{1 - e(x)}. \tag{12}$$

Applying the quotient rule $\phi_\tau = (\phi_{\mu_1} - \tau \cdot \phi_{\mu_0})/\mu_0$ and substituting nuisance estimates yields the DR-S/T pseudo-outcome:

$$\Gamma^{\text{DR-S/T}} = \hat{\tau}(X) + \frac{W(Y - \hat{\mu}_1(X))}{\hat{e}(X) \cdot \hat{\mu}_0(X)} - \frac{\hat{\tau}(X)(1 - W)(Y - \hat{\mu}_0(X))}{(1 - \hat{e}(X)) \cdot \hat{\mu}_0(X)} \tag{13}$$

where $\hat{\tau}(X) = \hat{\mu}_1(X)/\hat{\mu}_0(X)$. The first correction term adjusts for bias in $\hat{\mu}_1$, the second for bias in $\hat{\mu}_0$, both weighted by the inverse propensity.

4.1.1 Double Robustness of DR-S/T

The DR-S/T estimator achieves *classical double robustness*: it is consistent for $\tau(x)$ if either

(a) the propensity model is correct: $\hat{e}(x) = e(x)$, or

(b) the outcome models are correct: $\hat{\mu}_0(x) = \mu_0(x)$ and $\hat{\mu}_1(x) = \mu_1(x)$.

We now verify both cases. Writing $\Gamma^{\text{DR-S/T}} = A + B - C$ with $A = \hat{\tau}$, $B = W(Y - \hat{\mu}_1)/(\hat{e}\hat{\mu}_0)$, and $C = \hat{\tau}(1 - W)(Y - \hat{\mu}_0)/((1 - \hat{e})\hat{\mu}_0)$:

- **Case (a): $\hat{e} = e$ (correct propensity).**

  Taking conditional expectations given $X = x$:

$$\mathbb{E}[B \mid X{=}x] = \frac{\mu_1(x) - \hat{\mu}_1(x)}{\hat{\mu}_0(x)}, \tag{14}$$

$$\mathbb{E}[C \mid X{=}x] = \frac{\hat{\tau}(x)(\mu_0(x) - \hat{\mu}_0(x))}{\hat{\mu}_0(x)}. \tag{15}$$

  A first-order Taylor expansion of $\tau = \mu_1/\mu_0$ around $(\hat{\mu}_1, \hat{\mu}_0)$ gives:

$$\tau(x) = \hat{\tau}(x) + \frac{\mu_1(x) - \hat{\mu}_1(x)}{\hat{\mu}_0(x)} - \frac{\hat{\tau}(x)(\mu_0(x) - \hat{\mu}_0(x))}{\hat{\mu}_0(x)} + O_p(\|\hat{\mu} - \mu\|_2^2) \tag{16}$$

  Therefore $\mathbb{E}[\Gamma^{\text{DR-S/T}} \mid X{=}x] = \tau(x) + O_p(\|\hat{\mu} - \mu\|_2^2)$.

- **Case (b): $\hat{\mu}_0 = \mu_0$, $\hat{\mu}_1 = \mu_1$ (correct outcome models).**

  Then $A = \tau(x)$, and $\mathbb{E}[B \mid X{=}x] = \mathbb{E}[C \mid X{=}x] = 0$ since $Y - \mu_w(X)$ has conditional mean zero given (W, X). Thus $\mathbb{E}[\Gamma^{\text{DR-S/T}} \mid X{=}x] = \tau(x)$ exactly, regardless of the propensity estimate.

When both models are misspecified, the bias is $O_p(\|\hat{e} - e\|_2 \cdot \|\hat{\mu} - \mu\|_2)$, second-order in the nuisance estimation errors.

4.1.2 DR-S/T on Log Scale

When treatment effects span a wide range, direct-scale pseudo-outcomes exhibit heteroscedastic variance proportional to the effect magnitude. A log-scale formulation provides variance stabilization. Since $\log \tau(x) = \log \mu_1(x) - \log \mu_0(x)$ and the influence functions for $\log \mu_w$ are $\phi_{\log \mu_w} = \phi_{\mu_w}/\mu_w$, we obtain:

$$\Gamma^{\text{DR-S/T}}_{\log} = \log \hat{\mu}_1(X) - \log \hat{\mu}_0(X) + \frac{W(Y - \hat{\mu}_1(X))}{\hat{e}(X)\hat{\mu}_1(X)} - \frac{(1 - W)(Y - \hat{\mu}_0(X))}{(1 - \hat{e}(X))\hat{\mu}_0(X)} \tag{17}$$

The log-scale DR-S/T inherits the same double robustness property as the direct-scale version.

4.2 DR-Q: Augmenting the Q-Learner

While DR-S/T achieves classical double robustness, the Q-Learner parameterization leads to a weaker robustness structure. The Q-Learner identity (4) expresses $\tau(x)$ as a product:

$$\tau(x) = \underbrace{\frac{p(x)}{1 - p(x)}}_{A(x)} \cdot \underbrace{\frac{1 - e(x)}{e(x)}}_{B(x)} \tag{18}$$

We apply the product rule $\phi_\tau = B\phi_A + A\phi_B$. For $A(x) = p(x)/(1-p(x))$, the influence function on the converter subpopulation, lifted to the full population via $m(x)$, is $\phi_A = Y(W-p(x))/(m(x)(1-p(x))^2)$. For $B(x) = (1 - e(x))/e(x)$, the influence function is $\phi_B = -(W - e(x))/e(x)^2$. Substituting nuisance estimates yields the DR-Q pseudo-outcome:

$$\Gamma^{\text{DR-Q}} = \hat{\tau}(X) + \frac{(1 - \hat{e}(X)) \cdot Y(W - \hat{p}(X))}{\hat{e}(X) \cdot \hat{m}(X)(1 - \hat{p}(X))^2} - \frac{\hat{p}(X)(W - \hat{e}(X))}{(1 - \hat{p}(X))\hat{e}(X)^2} \tag{19}$$

where $\hat{\tau}(X) = \frac{\hat{p}(X)}{1-\hat{p}(X)} \cdot \frac{1-\hat{e}(X)}{\hat{e}(X)}$ is the plug-in Q-Learner estimate. Since both parameterizations target the same functional $\tau(x)$, the augmentation terms are algebraically equivalent to those of DR-S/T (13) after substitution of the identities $\mu_1 = pm/e$ and $\mu_0 = (1-p)m/(1-e)$.

4.2.1 Conditional Robustness of DR-Q

Unlike DR-S/T, the DR-Q estimator does *not* achieve classical double robustness: the propensity score $e(x)$ appears in *both* the plug-in estimate and the augmentation terms.

Concretely, the DR-Q estimator requires *exact* specification of the propensity score $e(x)$. Given $\hat{e}(x) = e(x)$ **exactly**, the estimator is consistent for $\tau(x)$ if either:

(a) the marginal conversion model is correct: $\hat{m}(x) = m(x)$, or

(b) the converter propensity model is correct: $\hat{p}(x) = p(x)$

If $\hat{e}(x) \neq e(x)$, the estimator is generally inconsistent regardless of the quality of $\hat{m}$ and $\hat{p}$.

We verify both cases. Writing

$$\Gamma^{\text{DR-Q}} = \underbrace{\hat{A}\hat{B}}_{\text{plug-in}} + \underbrace{\hat{B} \cdot \hat{\phi}_A}_{p\text{-correction}} + \underbrace{\hat{A} \cdot \hat{\phi}_B}_{e\text{-correction}} \tag{20}$$

where $\hat{A} = \hat{p}/(1-\hat{p})$, $\hat{B} = (1-\hat{e})/\hat{e}$, and $\hat{\phi}_A$, $\hat{\phi}_B$ are the estimated influence function contributions.

- **Case (a): $\hat{e} = e$ and $\hat{m} = m$ (correct propensity and marginal conversion).**

  With correct propensity, $\hat{B} = B$ and $\mathbb{E}[\hat{\phi}_B \mid X] = 0$. For the p-correction term, note that $\mathbb{E}[Y(W - \hat{p}(X)) \mid X{=}x] = m(x)(p(x) - \hat{p}(x))$, so with $\hat{m} = m$ the marginal conversion cancels:

  $$\mathbb{E}[\hat{B} \cdot \hat{\phi}_A \mid X{=}x] = B(x) \cdot \frac{p(x) - \hat{p}(x)}{(1-\hat{p}(x))^2} \tag{21}$$

  Using a first-order Taylor expansion of $A = p/(1-p)$ around $\hat{p}$:

  $$A(x) = \hat{A}(x) + \frac{p(x) - \hat{p}(x)}{(1-\hat{p}(x))^2} + O_p(\|\hat{p} - p\|_2^2) \tag{22}$$

  Therefore:

  $$\mathbb{E}[\Gamma^{\text{DR-Q}} \mid X{=}x] = \hat{A}(x)B(x) + B(x)\frac{p(x) - \hat{p}(x)}{(1-\hat{p}(x))^2} = \tau(x) + O_p(\|\hat{p} - p\|_2^2) \tag{23}$$

- **Case (b): $\hat{e} = e$ and $\hat{p} = p$ (correct propensity and converter propensity).**

  Then $\hat{A} = A$, $\hat{B} = B$, so the plug-in equals $\tau(x)$. Both correction terms have conditional mean zero: $\mathbb{E}[\hat{\phi}_A \mid X] = 0$ since $W - p(X)$ has mean zero given $(Y{=}1, X)$, and $\mathbb{E}[\hat{\phi}_B \mid X] = 0$ since $W - e(X)$ has mean zero given X. Thus $\mathbb{E}[\Gamma^{\text{DR-Q}} \mid X{=}x] = \tau(x)$ exactly.

4.2.2 DR-Q on Log Scale

Taking logarithms of the Q-Learner identity gives $\log \tau(x) = \text{logit}(p(x)) - \text{logit}(e(x))$, where $\text{logit}(u) = \log(u/(1-u))$. Applying the delta method with $\frac{d}{dp}\text{logit}(p) = 1/(p(1-p))$:

$$\Gamma^{\text{DR-Q}}_{\log} = \underbrace{\text{logit}(\hat{p}(X)) - \text{logit}(\hat{e}(X))}_{\text{plug-in}} + \underbrace{\frac{Y(W - \hat{p}(X))}{\hat{m}(X)\hat{p}(X)(1-\hat{p}(X))}}_{\text{correction for } p} - \underbrace{\frac{W - \hat{e}(X)}{\hat{e}(X)(1-\hat{e}(X))}}_{\text{correction for } e} \tag{24}$$

The log-scale DR-Q has the same conditional robustness properties as the direct-scale version (Section 4.2.1).

4.2.3 DR-Q-Simple Learner

The conditional robustness of DR-Q demands $\hat{e}(x) = e(x)$. Under this requirement, the e-augmentation term has conditional expectation zero and provides no bias correction. Analogous to the Q-Simple Learner (Section 3.1) this motivates a simplified construction that operates only on converters ($Y = 1$) and eliminates the need to estimate $m(x)$:

$$\Gamma^{\text{DR-Q-Simple}} = \hat{\tau}(X) + \frac{(1 - e(X)) \cdot (W - \hat{p}(X))}{e(X) \cdot (1 - \hat{p}(X))^2} \tag{25}$$

where $\hat{\tau}(X) = \frac{\hat{p}(X)}{1-\hat{p}(X)} \cdot \frac{1-e(X)}{e(X)}$ and $e(x)$ is the known propensity. Given exact $e(x)$:

$$\mathbb{E}[\Gamma^{\text{DR-Q-Simple}} \mid X = x, Y = 1] = \tau(x) + O_p(\|\hat{p} - p\|_2^2) \tag{26}$$

The DR-Q-Simple learner is particularly attractive for RCT applications: it eliminates the need to estimate $m(x)$, operates only on the $n_1 = \sum_i Y_i$ converters, and requires only a single nuisance function $\hat{p}(x)$. The simplification comes at a cost in asymptotic efficiency.

4.2.4 DR-Q-Simple on Log Scale

Applying the same log-scale construction as in DR-Q (Section 4.1.2) but restricting to converters and using known $e(x)$:

$$\Gamma_{\text{log}}^{\text{DR-Q-Simple}} = \underbrace{\text{logit}(\hat{p}(X)) - \text{logit}(e(X))}_{\text{plug-in}} + \underbrace{\frac{W - \hat{p}(X)}{\hat{p}(X)(1 - \hat{p}(X))}}_{\text{correction for } p} \tag{27}$$

This pseudo-outcome is computed only for observations with $Y = 1$ and inherits the same robustness property: consistency for $\log \tau(x)$ with bias $O_p(\|\hat{p} - p\|_2^2)$.

5 Empirical Evaluation

We now systematically evaluate which learner performs best under which conditions to give guidance on when to use which. Further, the theoretical analysis in Section 4 leads to three predictions we want to test. First, in observational settings DR-S and DR-T should outperform DR-Q: their classical double robustness tolerates propensity misspecification, while DR-Q's conditional robustness does not. Second, we expect the simple variants of the Q-Learner family to be slightly worse than their non-simple counterparts, but by how much is an open question. Third, log-scale DR variants should be more stable than their direct-scale counterparts when treatment effects vary substantially in magnitude.

5.1 Setup

We evaluate twelve learners: four plug-in methods (S, T, Q, Q-Simple) and eight doubly robust variants (DR-T, DR-T log, DR-S, DR-S log, DR-Q, DR-Q log, DR-Q-Simple, and DR-Q-Simple log). Q-Simple and DR-Q-Simple are only applicable to RCT datasets where propensity is known by design; observational benchmarks use ten learners.

Datasets. We select datasets to cover a range of sample sizes (n = 1.6k–687k), conversion rates (0.9%–68%), and treatment fractions (18%–85%). We separate RCT datasets, where propensity is known by design, from observational datasets, where propensity must be estimated and confounding is present. This separation reflects the fundamentally different estimation challenges and allows us to test the theoretical distinction between classical and conditional double robustness.

Table 1: RCT benchmark datasets. Propensity $e(x)$ is known by design. *Semi-synthetic: both potential outcomes observed, treatment simulated with $e = 0.5$. †10% random subsample for computational feasibility (full dataset: 13.9M rows).

Dataset	n	Conv.	Treat.	Domain
Criteo† [Diemert et al., 2018]	140k	4.7%	85.0%	E-commerce
Hillstrom (Visit) [Hillstrom, 2008]	64k	14.7%	66.8%	Email
Hillstrom (Conv.) [Hillstrom, 2008]	64k	0.9%	66.8%	Email
MegaFon [MegaFon, 2019]	600k	20.4%	50.1%	Telecom
X5 Retail [X5 Retail, 2020]	200k	62.0%	49.9%	Retail
Lenta [Lenta, 2021]	687k	5.9%	50.0%	Grocery
Twins* [Louizos et al., 2017]	71k	3.6%	50.0%	Neonatal

Table 2: Observational benchmark datasets. Propensity $e(x)$ must be estimated.

Dataset	n	Conv.	Treat.	Domain
LaLonde [LaLonde, 1986]	2.7k	60.0%	36.6%	Job training
RHC [Connors et al., 1996]	5.7k	68.5%	38.0%	Medical
Cattaneo [Cattaneo, 2010]	4.6k	6.0%	18.6%	Birthweight
NHEFS [Hernán and Robins, 2020]	1.6k	19.5%	26.3%	Smoking
JTPA [Abadie et al., 2002]	9.2k	50.0%	44.0%	Job training

Metrics. We evaluate two complementary metrics. The *Qini coefficient* measures ranking quality: how well a learner identifies individuals with the highest treatment effect. We adapt the normalized Qini coefficient [Radcliffe and Surry, 2011] to ratio-based effects by integrating the ratio rather than the difference:

$$\text{Qini}_{\text{ratio}} = \int_0^1 \frac{\hat{Y}^{(1)}(s)}{\hat{Y}^{(0)}(s)}\,\mathrm{d}s - \frac{Y^{(1)}}{Y^{(0)}},$$

where $\hat{Y}^{(w)}(s)$ is the mean outcome in arm w among the top-s fraction ranked by predicted $\hat{\tau}(x)$. Higher is better.

The *multiplicative calibration error* measures how accurately predicted effect sizes match realized effects. We compute the error on the log scale, where over- and under-prediction are treated symmetrically ($\log(0.8) = -\log(1.25)$), then exponentiate to obtain an interpretable multiplicative factor:

$$\text{CalError} = \exp\left(\frac{\sum_{b=1}^{B} n_b \left|\log \bar{\hat{\tau}}_b - \log \hat{\tau}_b^{\text{emp}}\right|}{\sum_{b=1}^{B} n_b}\right), \tag{28}$$

where $\bar{\hat{\tau}}_b$ is the mean predicted effect in bin b and $\hat{\tau}_b^{\text{emp}} = \bar{Y}_b^{(1)}/\bar{Y}_b^{(0)}$ is the empirical ratio within that bin ($B = 10$ equal-frequency bins). A CalError of 1.0× indicates perfect calibration; 1.5× means predictions are on average off by a factor of 1.5.

We regard Qini as more important than calibration: a badly ranking model cannot be recovered, while bad calibration can be restored. We demonstrate this in the simplest way below, using a log-linear transformation of $\hat{\tau}$, but envision more powerful ways to recover calibration as future work.

Comparison across datasets. Since absolute Qini values vary by orders of magnitude across datasets (e.g., MegaFon ≈ 1.3 vs. Lenta ≈ 0.02), a fixed absolute gap would be dominated by large-scale datasets. We therefore summarize learner performance using the *ratio to the best*

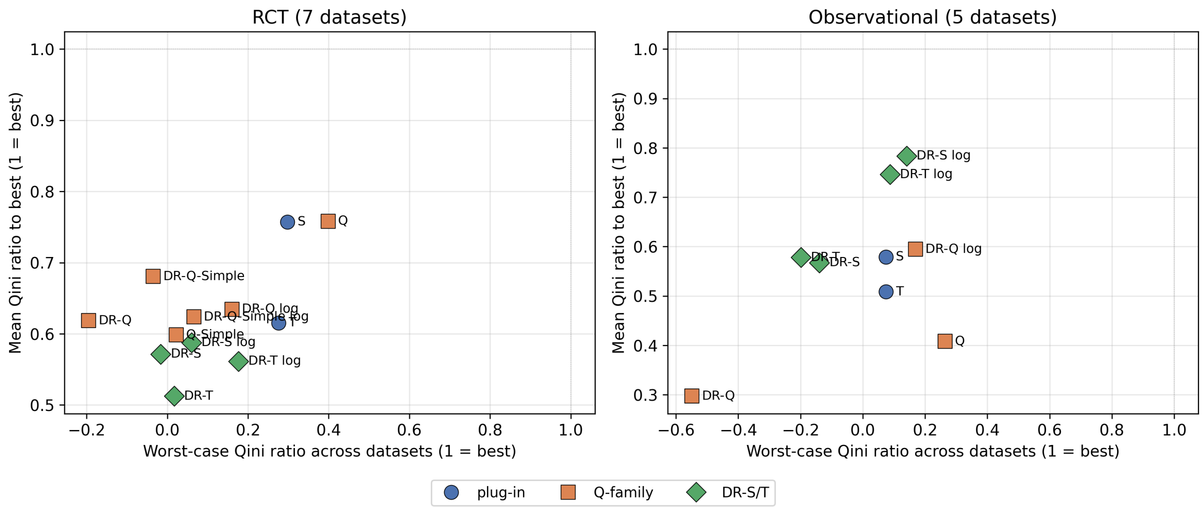

Figure 1: Mean Qini ratio to the best learner per dataset (1 = best) vs. worst-case Qini ratio across datasets (1 = best), for RCT (left) and observational (right) datasets. The ideal learner sits in the upper-right corner (high mean, high worst-case).

learner per dataset:

$$R^Q_{l,d} = \frac{Q_{l,d}}{\max_{l'} Q_{l',d}}. \tag{29}$$

This quantity is at most 1, with 1 indicating the best learner on that dataset; it can be negative when a learner's Qini itself is negative (i.e., worse-than-random targeting). Averaging $R^Q_{l,d}$ across datasets yields the mean Qini ratio: the average fraction of the dataset-optimal Qini that a learner attains. Analogously, for calibration we compute the ratio to the best-calibrated learner: $R^C_{l,d} = \mathrm{CalError}_{l,d}/\min_{l'} \mathrm{CalError}_{l',d}$ (always ≥ 1, with 1 indicating best calibration; a value of 2 means the learner's calibration error is twice that of the best-calibrated learner on that dataset). These ratios are invariant to dataset-specific scale, so a Lenta-sized Qini of 0.02 and a MegaFon-sized Qini of 1.3 contribute comparably to the cross-dataset average. To complement the mean with a worst-case view, we additionally report $R^Q_{\min} = \min_d R^Q_{l,d}$ (and analogously $R^C_{\max} = \max_d R^C_{l,d}$ for calibration): the worst per-dataset ratio a learner attains. A high mean paired with a high $R^Q_{\min}$ indicates a learner that is stable across datasets. Full per-dataset Qini values and standard errors across 100 seeds are in Appendix A.

Implementation. All methods use LightGBM [Ke et al., 2017] for nuisance and final-stage models with default hyperparameters. DR learners use 5-fold cross-fitting [Chernozhukov et al., 2018] to generate out-of-fold nuisance predictions. Final-stage regressors use a Poisson objective to enforce $\hat{\tau}(x) > 0$. To improve stability, we clip nuisance estimates and pseudo-outcomes as described in Appendix B. Results are averaged over 100 random seeds per dataset, with each seed determining the train/test split.

5.2 Ranking Analysis

We evaluate which learners most consistently achieve near-optimal ranking performance across datasets. Figure 1 summarizes the results using the ratio-to-best metric (29): learners closer to one with lower variance are more reliably competitive.

5.2.1 RCT Datasets

Table 3 (Appendix A) reports per-dataset Qini coefficients on the seven RCT datasets. **No single learner dominates**: seven datasets produce six different winners (DR-S on two datasets, DR-Q-Simple, DR-Q, T-Learner, Q-Learner, and DR-T log). In the cross-dataset ratio-to-best summary (Figure 1, left panel), the **Q-Learner and S-Learner are effectively tied** for the most consistently competitive method: both attain a mean Qini ratio of 0.76 of the dataset-optimal value, but the Q-Learner is markedly more stable across datasets: its worst-case ratio is $R^Q_{\text{min}} = 0.40$, compared to 0.30 for the S-Learner (which drops to that level on Lenta).

The broader pattern on RCT data is that **plug-in learners are more consistently competitive than their doubly robust counterparts**: the top two positions by mean Qini ratio are held by plug-ins (Q: 0.76, S: 0.76), while the best DR variant (DR-Q-Simple) reaches 0.68 and every other DR variant falls below 0.64. On the four RCTs where a DR variant is the outright winner (H(Vis), H(Conv), Criteo, X5), the best plug-in attains between 84% and 96% of the DR winner's Qini, so even in the cases where DR wins, plug-ins follow very closely. This matches the theoretical expectation: when propensity is known by design and no confounding can arise, plug-in learners can correctly identify the CATE. The DR augmentation then trades a small bias-correction benefit against added estimation variance from the nuisance correction term. While a DR learner sometimes still comes out on top, our results suggest that the additional variance often dominates the bias reduction.

Q-Learner vs. Q-Simple. The Q-Learner (mean Qini ratio 0.76) outperforms Q-Simple (0.60), but on the raw Qini scale the advantage is small (mean 0.37 vs. 0.36) and almost entirely driven by a single dataset: on Lenta, Q-Simple's Qini collapses to 0.002 while the Q-Learner reaches 0.089, so the ratio metric magnifies what is in absolute terms a small per-dataset difference. On the other six RCT datasets Q and Q-Simple are within 0.03 Qini of each other. Q-Simple therefore remains a competitive, computationally cheaper alternative when the raw Qini is the quantity of interest.

5.2.2 Observational Datasets

On observational data, where propensity must be estimated, the result reverses: DR learners consistently outperform plug-in learners. Table 4 (Appendix A) reports per-dataset Qini coefficients on the five observational datasets. The log-scale DR learners occupy the top three positions: **DR-S log** (0.78), DR-T log (0.75), DR-Q log (0.60). All three plug-in learners fall below the log-scale DR group: S (0.58), T (0.51), Q (0.41). The direct-scale DR learners are intermediate between the log-scale DR group and the plug-ins for DR-S (0.57) and DR-T (0.58), and below all other learners for DR-Q (0.30).

This confirms the first theoretical prediction from Section 5: outcome-regression-based DR learners with classical double robustness (DR-S/T) outperform both plug-in methods and DR-Q in observational settings once the log transform is applied. The advantage of DR-S over DR-T is small in this regime (log-scale: 0.78 vs. 0.75; direct-scale: 0.57 vs. 0.58), suggesting that the regularisation benefit of joint outcome modelling is largely absorbed once the log transform stabilises the pseudo-outcomes on the small observational datasets ($n = 1.6\text{k}$–9.2k).

Conditional vs. classical robustness. DR-Q direct produces negative Qini on both RHC (-0.106) and NHEFS (-0.082), with a mean ratio of 0.30 and a worst-case ratio of $R^Q_{\text{min}} = -0.55$ on RHC—the largest failure across all learners in the panel. However, attributing this entirely to conditional robustness would be misleading: the direct scale introduces additional variance that amplifies the problem. The cleaner test is DR-Q log (mean ratio 0.60), which isolates the robustness penalty by removing direct-scale variance. DR-Q log still lags DR-S log (0.78) and

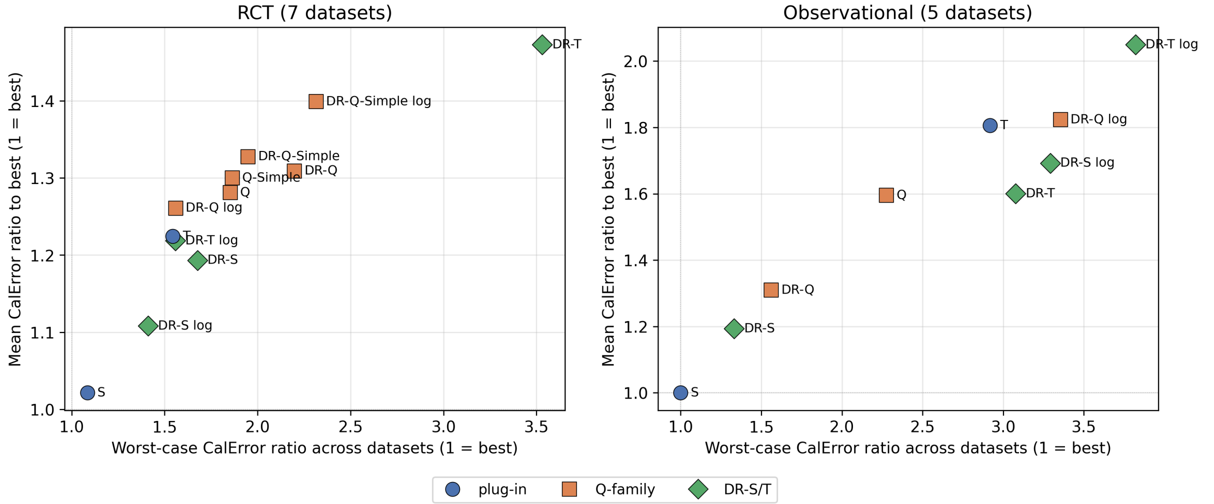

Figure 2: Mean calibration-error ratio to the best-calibrated learner per dataset (1 = best) vs. worst-case calibration-error ratio across datasets (1 = best), for RCT (left) and observational (right) datasets. Each learner at its best calibration configuration. The ideal learner sits at the bottom-left corner (low mean, low worst-case).

DR-T log (0.75), confirming that conditional robustness is a genuine disadvantage in observational settings, but the gap is moderate rather than catastrophic.

Log vs. direct scale. Across all DR learners, log-scale variants consistently outperform their direct-scale counterparts on observational data (DR-S log 0.78 vs. DR-S 0.57; DR-T log 0.75 vs. DR-T 0.58; DR-Q log 0.60 vs. DR-Q 0.30). The variance-stabilising reparametrisation is particularly beneficial on smaller datasets where direct-scale pseudo-outcomes can produce extreme values.

5.3 Calibration Analysis

Direct-scale DR pseudo-outcomes introduce a large systematic scale bias. To exemplify how calibration can be restored in hindsight we apply a rank-preserving log-linear recalibration ($\log \hat{\tau}_{\text{cal}} = \hat{a} + \hat{b} \log \hat{\tau}$, fit on bin-level means), which reduces calibration error for direct-scale DR learners (DR-Q: -27%, DR-T: -22%, DR-S: -10%) but worsens it for log-scale DR and plug-in methods (Table 7, Appendix A). Here, other methods must be applied, which is beyond the scope of the paper.

Figure 2 shows the calibration-error ratio across datasets. On RCT data, the **S-Learner** dominates calibration (mean ratio 1.02, $R^C_{\max} = 1.08$), followed by DR-S log (1.11) and DR-S (1.19). DR learners on direct scale remain poorly calibrated even after recalibration. On observational data, the pattern is similar: the S-Learner is the best-calibrated learner on every single observational dataset (mean ratio 1.00, $R^C_{\max} = 1.00$). The log-scale DR variants that dominate ranking are among the worst-calibrated learners: DR-T log reaches a mean ratio of 2.05 and DR-S log 1.69, substantially worse than their direct-scale counterparts (DR-T: 1.60, DR-S: 1.19). The ranking advantage of log-scale DR thus comes at the cost of poor calibration; post-hoc recalibration would be needed before using these models for score-level decisions.

5.4 Summary

1. **RCT: plug-in learners suffice.** On RCT data no single learner dominates, but the Q-Learner and S-Learner are the most consistently competitive methods, while every DR

variant lags behind. Even on datasets where a DR variant wins, the best plug-in follows closely. For practitioners who want a lighter alternative, Q-Simple is nearly as competitive as the Q-Learner on most RCT datasets and computationally cheaper.

2. **Observational: log-scale DR is necessary.** When propensity must be estimated, the picture reverses. Log-scale DR learners dominate, while plug-in learners fall well behind and direct-scale DR lags its log-scale counterparts. The log transform is the decisive ingredient. DR-Q direct performs poorly in this setting, producing negative Qini on two datasets; the log-scale variant fares better, but conditional robustness remains a genuine disadvantage when exact propensity knowledge is unavailable.

3. **Calibration: S-Learner is universally best.** The S-Learner dominates calibration in both settings, followed by other plug-in learners. As a proof of concept, we show that post-hoc log-linear recalibration can improve calibration without affecting Qini, demonstrating that bad calibration is recoverable in principle.

Limitations. Our evaluation uses binary outcomes, a single base learner (LightGBM with default hyperparameters), and fixed clipping thresholds. Results may differ with learner-specific hyperparameter tuning, other model classes, or continuous outcomes. The observational datasets are small ($n = 1.6\text{k}$–9.2k), limiting the precision of comparisons. Without access to true CATEs, we evaluate ranking and calibration but cannot assess pointwise estimation accuracy.

6 Conclusion

We introduced the Q-Learner, a meta-learner for ratio-based CATEs that decomposes $\tau(x)$ into a product of odds ratios via a Bayes-rule identity, reducing estimation to two binary classification tasks that directly optimize the treatment effect estimate. We derived doubly robust augmentations for both outcome-regression-based (DR-S/T) and Q-Learner-based (DR-Q) ratio estimators, and characterized their distinct robustness properties: classical double robustness for DR-S/T versus conditional robustness for DR-Q.

Our benchmark across twelve datasets and 100 seeds per dataset yields a consistent picture. In RCT settings with known propensity, plug-in learners prove sufficient: the Q-Learner and S-Learner are jointly the most consistent methods, while every DR variant lags behind. Even on the four RCT datasets where a DR variant wins, the best plug-in follows closely. For practitioners seeking a lighter alternative, Q-Simple is nearly as competitive as the Q-Learner on most RCT datasets and computationally cheaper. In observational settings, the picture reverses: the log-scale DR learners introduced here clearly dominate, confirming that classical double robustness and log-scale variance stabilisation are both necessary when propensity must be estimated. DR-Q direct produces worse-than-random targeting on two datasets; the log-scale variant fares better, but conditional robustness remains a genuine disadvantage when propensity must be estimated. Across both settings, the S-Learner is the best-calibrated method.

These results suggest a simple decision rule: use plug-in learners (the Q- or S-Learner) as the default on RCTs, falling back to Q-Simple when computational efficiency matters; use DR-S log when ranking is the primary objective in observational settings. Only when the absolute best ranking on a specific dataset is required should additional methods be evaluated: since no single learner dominates, all candidates should be tested in that case. Confidence intervals for ratio-based CATEs, building on the influence functions derived here, remain an open direction, as does extending the framework to continuous and time-to-event outcomes. Investigating whether the Q-Learner's consistency advantage grows with learner-specific hyperparameter tuning is an important practical question, since its component classifiers directly optimise $\hat{\tau}(x)$.

Code and data availability. All code, dataset loaders, and scripts to reproduce the experiments in this paper are available at `https://github.com/michaelfuchs90/ratiobasedcate`. The repository includes the 25,800-row benchmark CSV, the notebook that generates all figures and tables, and a README with replication instructions.

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References

Abadie, A., Angrist, J., and Imbens, G. (2002). Instrumental variables estimates of the effect of subsidized training on the quantiles of trainee earnings. *Econometrica*, 70(1):91–117.

Bickel, P. J., Klaassen, C. A., Ritov, Y., and Wellner, J. A. (1993). *Efficient and Adaptive Estimation for Semiparametric Models*. Springer.

Cattaneo, M. D. (2010). Efficient semiparametric estimation of multi-valued treatment effects under ignorability. *Journal of Econometrics*, 155(2):138–154.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Connors, A. F., Speroff, T., Dawson, N. V., et al. (1996). The effectiveness of right heart catheterization in the initial care of critically ill patients. *JAMA*, 276(11):889–897.

Diemert, E., Betlei, A., Renaudin, C., and Amini, M. R. (2018). A large scale benchmark for uplift modeling. *Proceedings of AdKDD & TargetAd, KDD*.

Boughdiri, A., Josse, J., and Scornet, E. (2025). Quantifying treatment effects: Estimating risk ratios via observational studies. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Hernán, M. A. and Robins, J. M. (2020). *Causal Inference: What If*. Chapman & Hall/CRC.

Hillstrom, K. (2008). The MineThatData E-Mail Analytics And Data Mining Challenge. *MineThatData Blog*.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS*, 30.

Kennedy, E. H. (2022). Semiparametric doubly robust targeted double machine learning: a review. *arXiv:2203.06469*.

Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *PNAS*, 116(10):4156–4165.

LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *The American Economic Review*, 76(4):604–620.

Lenta Uplift Modeling Dataset (2021). `https://sklift.readthedocs.io/en/latest/datasets.html`.

Louizos, C., Shalit, U., Mooij, J. M., Sontag, D., Zemel, R., and Welling, M. (2017). Causal effect inference with deep latent-variable models. *NeurIPS*, 30.

MegaFon Uplift Competition (2019). *Open Data Science Community*.

Nie, X. and Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319.

Robins, J. M., Rotnitzky, A., and Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866.

Radcliffe, N. J. and Surry, P. D. (2011). Real-world uplift modelling with significance-based uplift trees. *White Paper, Stochastic Solutions*.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701.

van der Vaart, A. W. (2000). *Asymptotic Statistics*. Cambridge University Press.

X5 Retail Hero Uplift Competition (2020). `https://retailhero.ai/`.

Zou, G. (2004). A modified Poisson regression approach to prospective studies with binary data. *American Journal of Epidemiology*, 159(7):702–706.

Andersen, L. W. (2021). Absolute vs. relative effects—implications for subgroup analyses. *Trials*, 22(1):50.

Athey, S., Tibshirani, J., and Wager, S. (2019). Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178.

Deeks, J. J. (2002). Issues in the selection of a summary statistic for meta-analysis of clinical trials with binary outcomes. *Statistics in Medicine*, 21(11):1575–1600.

Gao, Z. and Hastie, T. (2025). Estimating heterogeneous treatment effects for general responses. *Biometrics*, 81(4):ujaf162.

Schmid, C. H., Lau, J., McIntosh, M. W., and Cappelleri, J. C. (1998). An empirical study of the effect of the control rate as a predictor of treatment efficacy in meta-analysis of clinical trials. *Statistics in Medicine*, 17(17):1923–1942.

Shirvaikar, V., Storås, A., Lin, X., and Holmes, C. (2023). Targeting relative risk heterogeneity with causal forests. *arXiv preprint arXiv:2309.15793*.

Foster, D. J. and Syrgkanis, V. (2023). Orthogonal statistical learning. *Annals of Statistics*, 51(3). arXiv:1901.09036.

Greenland, S. (2004). Model-based estimation of relative risks and other epidemiologic measures in studies of common outcomes and in case-control studies. *American Journal of Epidemiology*, 160(4):301–305.

Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2). arXiv:2004.14497.

Richardson, T. S., Robins, J. M., and Wang, L. (2017). On modeling and estimation for the relative risk and risk difference. *Journal of the American Statistical Association*, 112(519):1121–1130.

Yadlowsky, S., Pellegrini, F., Lionetto, F., Braune, S., and Tian, L. (2021). Estimation and validation of ratio-based conditional average treatment effects using observational data. *Journal of the American Statistical Association*, 116(536):1946–1960.

A Full Benchmark Results

A.1 Qini on RCT Datasets

Table 3: Qini coefficient on RCT datasets (mean over 100 seeds; SE in parentheses). Best per column in **bold**. $R_{\min}$ is the worst-case per-dataset ratio $Q_{l,d}/\max_{l'} Q_{l',d}$ across datasets. Bottom row: mean across all learners per dataset.

Learner	H(Vis)	H(Conv)	Criteo	MegaFon	X5	Lenta	Twins	$\bar{Q}$	$R_{\min}$
S	0.435	0.371	0.334	1.424	0.073	0.027	0.087	**0.393**	0.298
	(0.016)	(0.065)	(0.007)	(0.005)	(0.001)	(0.005)	(0.006)		
T	0.321	0.181	0.263	**1.533**	0.055	0.025	0.076	0.351	0.276
	(0.015)	(0.049)	(0.007)	(0.005)	(0.001)	(0.004)	(0.008)		
Q	0.219	0.507	0.454	1.228	0.030	**0.089**	0.086	0.373	0.399
	(0.012)	(0.053)	(0.009)	(0.004)	(0.001)	(0.004)	(0.008)		
Q-Simple	0.215	0.516	0.427	1.236	0.025	0.002	0.083	0.358	0.022
	(0.012)	(0.051)	(0.007)	(0.004)	(0.001)	(0.003)	(0.008)		
DR-Q	0.356	0.580	**0.508**	1.184	0.070	−0.017	0.013	0.385	−0.195
	(0.016)	(0.055)	(0.009)	(0.004)	(0.001)	(0.005)	(0.009)		
DR-Q log	0.268	0.514	0.193	1.263	0.058	0.014	0.085	0.342	0.160
	(0.015)	(0.054)	(0.004)	(0.004)	(0.001)	(0.003)	(0.006)		
DR-Q-Simple	0.336	**0.615**	0.461	1.145	0.043	−0.003	0.080	0.383	−0.035
	(0.015)	(0.061)	(0.009)	(0.004)	(0.001)	(0.003)	(0.006)		
DR-Q-Simple log	0.243	0.600	0.449	1.229	0.026	0.006	0.073	0.375	0.066
	(0.014)	(0.049)	(0.008)	(0.004)	(0.001)	(0.003)	(0.011)		
DR-T	0.438	0.010	0.331	1.236	0.075	0.016	0.003	0.301	0.017
	(0.017)	(0.035)	(0.008)	(0.004)	(0.001)	(0.005)	(0.010)		
DR-T log	0.319	0.109	0.157	1.253	0.059	0.016	**0.092**	0.286	0.177
	(0.014)	(0.040)	(0.003)	(0.004)	(0.001)	(0.004)	(0.006)		
DR-S	**0.473**	0.321	0.343	1.195	**0.076**	0.004	−0.002	0.344	−0.016
	(0.017)	(0.053)	(0.007)	(0.004)	(0.002)	(0.005)	(0.010)		
DR-S log	0.330	0.361	0.102	1.259	0.058	0.005	0.091	0.315	0.060
	(0.015)	(0.060)	(0.002)	(0.004)	(0.001)	(0.003)	(0.006)		
Dataset mean	0.329	0.391	0.335	1.265	0.054	0.015	0.064		

A.2 Qini on Observational Datasets

Table 4: Qini coefficient on observational datasets (mean over 100 seeds; SE in parentheses). Best per column in **bold**. $R_{\min}$ is the worst-case per-dataset ratio $Q_{l,d}/\max_{l'} Q_{l',d}$ across datasets. Bottom row: mean across all learners per dataset.

Learner	LaLonde	RHC	Cattaneo	NHEFS	JTPA	$\bar{Q}$	$R_{\min}$
S	0.153	0.014	1.124	0.132	0.032	0.291	0.074
	(0.022)	(0.005)	(0.096)	(0.047)	(0.006)		
T	0.146	0.014	1.098	0.129	0.013	0.280	0.074
	(0.020)	(0.005)	(0.098)	(0.048)	(0.006)		
Q	0.083	0.061	0.637	0.111	0.019	0.182	0.264
	(0.019)	(0.008)	(0.107)	(0.041)	(0.006)		
DR-Q	0.124	−0.106	0.726	−0.082	**0.071**	0.147	−0.549
	(0.019)	(0.011)	(0.091)	(0.043)	(0.008)		
DR-Q log	0.101	0.087	1.040	0.229	0.012	0.294	0.169
	(0.022)	(0.014)	(0.115)	(0.045)	(0.006)		
DR-T	**0.154**	0.098	0.922	−0.050	0.063	0.237	−0.199
	(0.023)	(0.011)	(0.099)	(0.038)	(0.008)		
DR-T log	0.133	0.193	1.126	0.234	0.006	0.338	0.087
	(0.022)	(0.015)	(0.099)	(0.036)	(0.006)		
DR-S	0.133	0.098	0.866	−0.035	0.067	0.226	−0.139
	(0.021)	(0.011)	(0.081)	(0.035)	(0.008)		
DR-S log	0.120	**0.193**	**1.313**	**0.253**	0.010	**0.378**	0.141
	(0.022)	(0.015)	(0.108)	(0.041)	(0.006)		
Dataset mean	0.127	0.072	0.984	0.102	0.033		

A.3 Calibration

Table 5: CalError on RCT datasets (best of base/calibrated, mean over 100 seeds; SE in parentheses). Best per column in **bold**. $R_{\max}$ is the worst-case per-dataset ratio $\mathrm{CE}_{l,d}/\min_{l'}\mathrm{CE}_{l',d}$ across datasets. Bottom row: mean across all learners per dataset.

Learner	H(Vis)	H(Conv)	Criteo	MegaFon	X5	Lenta	Twins	$\overline{\mathrm{CE}}$	$R_{\max}$
S	**1.224**	2.316	1.110	1.076	**1.024**	**1.113**	1.397	**1.323**	1.084
	(0.006)	(0.043)	(0.004)	(0.002)	(0.001)	(0.003)	(0.012)		
T	1.490	3.295	1.211	1.103	1.057	1.322	1.961	1.634	1.543
	(0.007)	(0.069)	(0.005)	(0.001)	(0.001)	(0.004)	(0.018)		
Q	1.507	3.958	1.130	1.125	1.078	1.290	2.137	1.746	1.853
	(0.008)	(0.104)	(0.003)	(0.001)	(0.001)	(0.004)	(0.023)		
Q-Simple	1.514	3.979	1.153	1.130	1.075	1.387	2.148	1.770	1.863
	(0.008)	(0.100)	(0.005)	(0.001)	(0.001)	(0.005)	(0.024)		
DR-Q	1.525	4.660	1.142	1.077	1.046	1.465	1.757	1.810	2.182
	(0.010)	(0.141)	(0.005)	(0.001)	(0.001)	(0.005)	(0.021)		
DR-Q log	1.890	3.240	1.157	1.098	1.054	1.422	1.827	1.670	1.545
	(0.011)	(0.085)	(0.003)	(0.002)	(0.001)	(0.006)	(0.020)		
DR-Q-Simple	1.613	4.147	1.251	1.087	1.091	1.849	1.536	1.796	1.942
	(0.009)	(0.136)	(0.009)	(0.001)	(0.001)	(0.013)	(0.013)		
DR-Q-Simple log	1.621	4.903	1.184	1.115	1.074	1.398	2.328	1.946	2.296
	(0.009)	(0.119)	(0.004)	(0.001)	(0.001)	(0.004)	(0.027)		
DR-T	1.512	7.541	1.115	**1.048**	1.047	1.487	1.594	2.192	3.531
	(0.011)	(0.227)	(0.005)	(0.001)	(0.001)	(0.006)	(0.014)		
DR-T log	1.890	2.784	1.234	1.092	1.055	1.457	1.583	1.585	1.545
	(0.011)	(0.060)	(0.006)	(0.001)	(0.001)	(0.006)	(0.015)		
DR-S	1.373	3.568	**1.101**	1.064	1.045	1.517	1.573	1.606	1.671
	(0.008)	(0.081)	(0.003)	(0.001)	(0.001)	(0.006)	(0.014)		
DR-S log	1.719	**2.136**	1.175	1.098	1.053	1.337	**1.363**	1.411	1.405
	(0.010)	(0.041)	(0.004)	(0.002)	(0.001)	(0.005)	(0.011)		
Dataset mean	1.573	3.877	1.164	1.093	1.058	1.420	1.767		

Table 6: CalError on observational datasets (best of base/calibrated, mean over 100 seeds; SE in parentheses). Best per column in **bold**. $R_{\max}$ is the worst-case per-dataset ratio $\mathrm{CE}_{l,d}/\min_{l'}\mathrm{CE}_{l',d}$ across datasets. Bottom row: mean across all learners per dataset.

Learner	LaLonde	RHC	Cattaneo	NHEFS	JTPA	$\overline{\mathrm{CE}}$	$R_{\max}$
S	**1.573**	–	**2.156**	**1.853**	**1.191**	**1.693**	1.000
	(0.019)	(–)	(0.042)	(0.034)	(0.005)		
T	1.648	–	6.293	3.931	1.362	3.308	2.919
	(0.020)	(–)	(0.161)	(0.104)	(0.006)		
Q	1.818	1.776	3.769	4.203	1.417	2.597	2.269
	(0.029)	(0.010)	(0.123)	(0.109)	(0.007)		
DR-Q	1.780	1.273	3.050	2.651	1.391	2.029	1.431
	(0.022)	(0.012)	(0.075)	(0.050)	(0.007)		
DR-Q log	2.044	1.528	3.995	6.086	1.481	3.027	3.285
	(0.035)	(0.013)	(0.118)	(0.182)	(0.006)		
DR-T	1.736	1.126	5.263	2.930	1.410	2.493	2.441
	(0.023)	(0.006)	(0.149)	(0.065)	(0.007)		
DR-T log	2.089	1.239	5.757	7.092	1.492	3.534	3.828
	(0.036)	(0.008)	(0.158)	(0.221)	(0.006)		
DR-S	1.764	**1.126**	2.708	2.303	1.389	1.858	1.256
	(0.023)	(0.006)	(0.057)	(0.046)	(0.007)		
DR-S log	2.180	1.239	2.991	5.891	1.475	2.755	3.180
	(0.040)	(0.008)	(0.071)	(0.143)	(0.007)		
Dataset mean	1.848	1.330	3.998	4.104	1.401		

A.4 Recalibration

Table 7: Effect of post-hoc log-linear recalibration. DR-Q-Simple and Q-Simple: 7 RCT datasets only. Calibration is rank-preserving: Δ Qini $= 0$.

Learner	CalError (base)	CalError (cal.)	Δ
DR-Q	2.70×	1.97×	−27%
DR-Q log	2.42×	3.00×	+24%
DR-Q-Simple	2.30×	2.11×	−8%
DR-Q-Simple log	1.95×	3.28×	+68%
DR-T	3.17×	2.48×	−22%
DR-T log	2.42×	3.35×	+38%
DR-S	2.16×	1.95×	−10%
DR-S log	2.01×	2.73×	+36%
S-Learner	1.47×	2.06×	+40%
T-Learner	2.24×	4.86×	+117%
Q-Learner	2.12×	3.67×	+73%
Q-Simple	1.77×	3.83×	+116%

B Numerical Stability through Clipping

For numerical stability, we recommend the following clipping bounds as good starting points:

- **Propensity scores**: Clip $\hat{e}(x)$ to $[\epsilon_e, 1-\epsilon_e]$ with $\epsilon_e = 0.01$
- **Outcome probabilities**: Clip $\hat{\mu}_w(x)$ to $[\epsilon_\mu, 1-\epsilon_\mu]$ with $\epsilon_\mu = 0.001$
- **Marginal conversion**: Clip $\hat{m}(x)$ to $[\epsilon_m, 1]$ with $\epsilon_m = 0.001$
- **Converter propensity**: Clip $\hat{p}(x)$ to $[\epsilon_p, 1-\epsilon_p]$ with $\epsilon_p = 0.01$
- **Pseudo-outcomes**: Clip Γ^{DR} to $[\epsilon_\tau, 1/\epsilon_\tau]$ with $\epsilon_\tau = 0.01$
- **Log-scale pseudo-outcomes**: Clip $\Gamma^{\mathrm{DR}}_{\log}$ to $[-C, C]$ with $C = 10$

While these can serve as a good starting point, in practice we recommend treating the clipping parameters as hyperparameters subject to optimization.